

PROJECT SUMMARY REPORT

Social Media Engagement Analysis | Python Data Analysis

Tools: Python | Pandas | NumPy | Matplotlib | Seaborn | Plotly

1. Project Overview

This project analyses a social media dataset using Python to identify patterns in user engagement, content performance, audience demographics, and sentiment behaviour. The objective is to surface key drivers of engagement across post types, content categories, and user segments, and translate these findings into actionable business insights. The dataset covers users aged 13–64 across multiple countries and includes metrics such as post type, content category, engagement rate, likes, shares, comments, impressions, watch time, follower count, device type, and sentiment.

2. Data Preparation

The dataset was imported using Pandas and preprocessed to ensure quality and consistency. Missing values were handled appropriately, duplicate records were removed, and data types were corrected and standardised across all fields. Feature engineering was performed by creating a hashtag count variable to enrich the dataset for further analysis.

3. Data Exploration and Statistical Analysis

Descriptive statistics were generated across all numerical fields to profile engagement rate, likes, shares, comments, impressions, watch time, and follower count. A correlation analysis revealed predominantly weak associations between most variables, confirming that engagement is multifactorial. GroupBy operations segmented data by post type, content category, country, age group, device type, verification status, and sentiment to surface performance differences across key dimensions.

Key Statistical Findings

- Users ranged from 13 to 64 years with an average age of 38 years; highest engagement was among 24-year-olds, followed by users aged 55 and 51.
- Male users formed the largest gender group (36.98%), with an overall balanced gender distribution across the dataset.
- Verified accounts generated marginally higher average engagement than non-verified accounts.
- Follower counts were distributed similarly across all sentiment categories, confirming that audience size does not determine sentiment-based engagement.

4. Visualization Insights

- Impressions and engagement are weakly correlated — higher reach does not reliably drive more likes or interaction.
- Daily engagement rates fluctuated without a consistent trend, reflecting the variable nature of social media performance.
- Reels were posted most frequently but achieved the lowest average engagement; Videos were posted least but achieved the highest engagement and likes.
- The engagement rate distribution contained high-value outliers, indicating a small subset of posts achieved exceptional performance.
- The correlation heatmap confirmed weak inter-variable relationships, reinforcing that no single metric reliably predicts engagement.

- Watch time showed near-zero correlation with engagement rate — viewing duration alone does not drive interaction.
- Interactive visualisations revealed considerable variation in likes, impressions, and watch time, highlighting highly uneven content performance.
- Food content consistently outperformed other categories across chart types, while Travel content fell below average.

5. Final Insights

Content Performance

Video posts achieved the highest average engagement rate despite being the least frequently published format, while Reels — the most common post type — recorded the lowest engagement. Food was the best-performing content category, followed by Tech and Lifestyle; Travel received the lowest average engagement. Brazil recorded the highest average engagement rate by country, followed by Australia and France.

User Trends

The dataset reflects a diverse, multi-generational audience (ages 13–64, average 38). The highest engagement was observed among 24-year-old users, with users aged 55 and 51 also performing above average, indicating that engagement is not concentrated in younger demographics. Verified accounts generated slightly higher engagement than non-verified accounts, suggesting that perceived credibility may positively influence audience interaction.

Behavioral Insights

Time-of-day analysis could not be conducted as the dataset contains only posting dates and no time information. Mobile devices recorded the highest average watch time compared to Tablet and Desktop, consistent with mobile-first content consumption trends. Watch time showed almost no relationship with engagement rate, indicating that longer viewing duration does not necessarily translate to higher interaction.

Sentiment Analysis

Negative sentiment posts achieved the highest average engagement rate, followed by Neutral and Positive posts. Negative posts also received the most likes and shares, suggesting that emotionally provocative content drives stronger interaction. Neutral posts generated the highest average comment count, indicating they may encourage more discussion. Positive sentiment posts performed comparatively lower across most engagement metrics.

6. Conclusion

This project demonstrates the end-to-end application of Python-based data analysis to a social media dataset, encompassing data preparation, exploratory analysis, statistical examination, and multi-layered visualisation. The findings confirm that social media engagement is a complex, multifactorial outcome shaped by post format, content theme, geographic market, user demographics, device type, and sentiment — with no single metric acting as a reliable standalone predictor.

The analysis yields clear strategic implications: Video content delivers the strongest return despite low posting frequency; Food and Tech categories attract the highest audience interest; Brazil, Australia, and France represent high-engagement markets; and negative sentiment content, while counterintuitive, consistently drives stronger interaction. These insights provide a data-driven foundation for optimising content strategy, audience targeting, and platform engagement — demonstrating the core competencies of a professional Data Analyst.